



NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 3

CANDIDATE
NAME

CLASS

BIOLOGY

9816/01

Paper 1

September 2021

Candidates answer on the Question Paper.

2 hour 30 minutes

Additional Materials: Insert

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper

For Examiner's Use	
1	
2	
Section B	
Total	

This document consists of **11** printed pages and **1** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

1 You should read through the whole of the insert and **Question 1** carefully and then use the information provided to answer the questions. The insert includes **Fig. 1.1 – 1.3** and **Table 1.1 - 1.2.**

- (a) Suggest the role of cyanobacteria in the evolution of the circadian clock in some eukaryotic organisms.

primitive cyanobacteria evolved into the chloroplasts found in algae and plants;

Primitive eukaryotic cell engulfed a photosynthetic cyanobacteria;

Evolving eukaryotic cell gained energy conversion abilities while symbiotic cyanobacterium gained protection;

Retention of clock genes from cyanobacteria in chloroplast;

[3]

- (b) Identify the photosynthetic pathway of plants that is likely to be most affected by the circadian clock and explain the importance of circadian rhythm in these plants.

Identified pathway to be that of CAM plants;

Open their stomata during the night and close them during the day to minimise water loss through transpiration;

PEP carboxylase carries out carbon fixation at night when stomata is open and combines CO₂ to PEP, thereafter converting oxaloacetate to malate;

Malate releases CO₂ in the day which allows for high concentration of CO₂ around rubisco, therefore minimising photorespiration;

[3]

(c) With reference to Fig. 1.1 in the insert, explain the principles of homeostasis.

Stimulus (input) – a detectable change in the internal or external environment, such as a change in body temperature above 37

Receptor (detector) – temp sensitive nerve endings detects stimuli (change in temperature) and sends information to the thermoregulatory control centre;

Control centre (regulator) – determines the range within which a variable is to be maintained / determines the set point (37).

Coordinates information received from the receptors and sends out instructions to the effectors to carry out response to correct the deviation. Brain receives signals sent by nerve endings and sends signals to target organs (smooth muscle and sweat glands)

Effector (output) – Acts in response to control centre's instructions and acts to restore variable to set point. Smooth muscles in skin blood vessel and skeletal muscles contract, causing dilation of blood vessels and sweating, which increases heat loss.

Negative feedback – the response of the system then feedbacks to decrease the magnitude of the stimulus that triggered the events, thus returning the variable to set point;

[4]

(d) With reference to Fig. 1.2 in the insert,

(i) explain how the disruption of the circadian clock could result in depression.

Reduced levels of serotonin neurotransmitter released due to production / release at the wrong time;

Less receptors are bound, therefore less ligand-gated cation channels open, less influx of cations,

EPSP generated via depolarisation is not sufficient to reach threshold / EPSP not generated, hence no action potential generated, inability to transmit nerve impulse so not able to stimulate feelings of joy;

[3]

(ii) suggest reasons why nerve impulses only travel in one direction across the synapse.

Synaptic vesicles containing serotonin neurotransmitters only found at pre-synaptic neurone ® membrane;

Mitochondria for ATP synthesis for any energy-driven processes, e.g. active transport of serotonin, only found at pre-synaptic neurone; @enzyme required for synthesis of serotonin

Receptors for serotonin is only found at the post-synaptic membrane;

[2]

- (e) With reference to Fig. 1.3 in the insert and your knowledge,
 (i) describe the characteristics of a normal human prion protein.

Disordered N-terminal which is rich in positively charged amino acids and contains four octapeptide repeats;

C-terminal domain has a globular structure with three alpha helices and two short beta strands forming a short antiparallel beta sheet;

Glycosylphosphatidylinositol (GPI) anchor allowing the protein to attach to the cell surface membrane;

N-linked glycosylation / Addition of oligosaccharides at Asn residues at C terminus;

Protein is soluble and susceptible to degradation;

[3]

- (ii) Explain how prions are involved in the pathogenesis of human prion diseases.

Misfolding of PrP^c to PrP^{Sc} which has a higher percentage of beta pleated sheets and therefore become more stable, more resistant to proteolysis by proteases;

Misfolded proteins can trigger normal prions to change into misfolded shape, creating a positive feedback loop,

therefore resulting in aggregation into amyloid fibrils which destroys brain tissues;

[2]

- (f) A t-test is conducted and the student proposed the null hypothesis:

there is no difference between the sleep hours of control group and CJD group.

With reference to the results shown in Table 1.1 and the formulae provided below, calculate the value of t and the number of degrees of freedom.

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}} \quad v = n_1 + n_2 - 2$$

key to symbols

s = standard deviation

$\bar{x}$ = mean

n = sample size (number of observations)

v = degrees of freedom

Show your working.

$$t = |9.80 - 6.67| / \sqrt{10.82/10 + 8.82/12}$$

$$v = 10 + 12 - 2 \\ = 20$$

Value of t = 2.32

Number of degrees of freedom = 20

[1]

- (g) Use Table 1.2 and your answers to (f) to decide whether the null hypothesis suggested by the student should be accepted or rejected.

Explain your answer.

Calculated t test is greater than critical value at $p=0.05$ (2.09)

0.01 < p-value < 0.05, therefore reject null hypothesis.

[2]

- (h) Comment on the validity of the study.

Sample not representative of the population (e.g. gender not accounted for);

Should record data across more than one night and measure mean sleep hours;

Mean age being the same does not mean that the age of individuals in the 2 separate groups are the same.

The control group is comprised of non-CJD individuals who may still have other underlying health conditions which affect sleep hours;

[2]

[Total: 25]

2 “Natural selection is the mechanism of evolution”.

Discuss to what extent this statement is valid.

[25]

Introduction

- Definition: Evolution is the genetic changes that occur in a population of organisms overtime.
 - Microevolution – Changes in allele frequencies within a population
 - Macroevolution – Transcends the boundaries of a single species and involves the formation of new species from a common ancestor over many generations
- Definition of Fitness:

The success of an individual in leaving descendants to the next generation. It encompasses fitness components such as survival to maturity and mating success.

Natural selection

- Definition, description, examples to substantiate
- Inheritable variation occurs in individuals in a population leading to different survival abilities, therefore different fitness.
 - Possible sources: spontaneous random mutations, crossing over of homologous chromosomes, random fertilisation of gametes
- Ref. to essential features of Darwin’s theory of evolution by natural selection
 - Overproduction of offspring
 - Constancy of numbers
 - Variation in traits between individuals within a population
 - Struggle for survival and survival of the fittest (ref to competition for limited resources and therefore the individuals who are better adapted will survive and reproduce)
 - Like produces like (ref to individuals who survive and reproduce will pass on the beneficial alleles to the offspring, therefore resulting in increase in allelic frequency)
- Ref. to speciation processes in which natural selection contributes to the accumulation of differences between isolated populations (e.g. allopatric and sympatric speciation)
- Ref. to divergent evolution in which related species develop different adaptations to different selective pressures
 - Ref to NS in adaptive radiation where an ancestral species diversifies into a number of descendant species to occupy a broader range of ecological niches (description + substantiate with examples)
 - Ref to NS giving rise to ring species which allows for accumulation of differences between the populations as geographic distance increases (description + substantiate with examples)
- Ref. to convergent evolution of unrelated species developing similar adaptations, leading to the evolution of analogous structures
- Formation of new species through polyploidy, hybridisation, introgression
 - Definition + description + example
 - Ref to formation of offspring which have complete reproductive isolation from their ancestors
 - Ref to possible increased fitness due to hybrid vigour / more gene copies, resulting in increased gene product

Disagreement with “fittest” in NS

- Ref. to selection of individuals which are able to survive, not restricted to the best adapted – “fit enough” rather than “fittest”
- Ref. to what is considered as being fit changing as selective pressure changes
- Relevant examples – selection of individuals with sickle cell trait in places with malaria

Sexual selection

- Definition, description, examples for substantiation
 - Form of selection caused by competition for mates among individuals of the same sex
 - Ref to evolution of traits that can decrease survival if the reproductive advantage compensates for it, leading to an increase in male's overall fitness
 - Ref to intrasexual selection (competition for access to mates) and intersexual selection (mate choice, competition for attention from the other sex)
 - Ref. to sexual dimorphism due to sexual selection acting more strongly on males than females due to higher reproductive variance among males
 - Ref. to sexual selection being more pronounced when selective pressure (natural selection) is not as strong

Neutral theory of molecular evolution

- Contends that at the molecular level, most evolutionary changes and polymorphisms within species are not caused by natural selection but by random genetic drift
- Many molecular changes are selectively neutral, i.e. neither adaptive nor detrimental
- E.g. silent mutations due to degeneracy of the genetic code

Genetic drift

- Definition: process where chance events cause allelic frequencies to fluctuate in small populations
 - Ref to founders' effect and bottleneck event (description and examples to substantiate)

Artificial Selection

- Definition: process by which humans choose individual organisms with certain desirable phenotypic trait values for breeding
- Ref to distinction from NS due to selection applied by humans in order to produce genetic change
- E.g. domesticated species such as dogs for pets, crop plants (bigger fruits) and livestock (increased meat and milk production)

Communication (2m)

Mark	Descriptors
	Candidate has
2	Organised and presented information clearly and used correct terminology in appropriate contexts
1	Attempted to organize material and use some correct terminology, so that with re-reading the meaning becomes apparent
0	Presented an unstructured answer with poor use of terminology

Argumentation (3m)

Mark	Descriptors
	Candidate has
3	Developed and sustained its coherent argument throughout the essay leading to an appropriate conclusion showing insight
2	Introduced an argument and partially developed it, so that some coherence is shown in the essay
1	Shown evidence of an argument, with little development
0	Shown no evidence of argumentation

Section B

Answer **one** question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 3** Discuss the roles of B cells in immune response and immune tolerance. [25]
- 4** Discuss the role of enzymes in biological processes and biotechnological applications. [25]

Introduction

Blood stem cells, more specifically lymphoid stem cells divide and differentiate to give rise to B cells

B cells are produced in bone marrow and mature / develop in bone marrow.

Immune Response (12 max)

1. B cells are involved in humoral immunity of the adaptive (specific) immunity, which eliminate extracellular pathogen.
2. Somatic recombination:
 - Ref to mechanism of somatic recombination
 - Importance of somatic recombination: give rise to great diversity of BCR that can recognise wide range of foreign antigens (and self antigens)
3. Ref to activation of B cells (4m):
4. Ref to mechanism of somatic hypermutation & class switching:
5. Role of antibodies in humoral immunity (2m):
6. Ref to Structure of antibodies(3m) :
7. Role of Memory cells (2m):

Immune Tolerance (8 max)

1. Importance of immune tolerance – idea of unresponsive to self antigens / immune system does not mount response against self antigens / normal tissues of the body @ reverse (autoimmunity)
2. Ref to During development of B cell in bone marrow, immature B cells are tested for autoreactivity before they leave bone marrow.
3. Ref to location of central tolerance and peripheral tolerance
4. Central tolerance:
 - **Receptor editing**
 - **Clonal Deletion**

5. Peripheral Tolerance
 - **Clonal deletion**
 - **Anergy**
6. Ref to autoimmune diseases

Discuss the role of enzymes in biological processes and biotechnological applications.

1. Definition of Enzymes (2m): structure & function
2. Properties of enzymes (3m)
3. Specific examples of enzymes and their importance
4. Ref to DNA replication – DNA helicase / DNA topoisomerase / DNA polymerase
5. Ref to Protein synthesis – aminoacyl synthetase, peptidyl transferase
6. Ref to cell signalling: Kinases / Phosphatases
7. Ref to telomerase
8. Hydrolytic enzymes in lysosomes
9. Ribozymes
10. Restriction enzymes (natural function in bacterial cells)
11. Biotechnological applications: recombinant DNA technology → Restriction enzymes, DNA ligase.
12. cDNA libraries – reverse transcriptase and DNA polymerase
13. PCR: Taq polymerase